These instructions reproduce the exact experimental results reported in the accompanying report. Following these steps will regenerate all metrics, statistical tests, plots and cross-project results.

Software Environment

- *Python 3.8+*
- See *requirements.pdf* for exact library versions

Step 1 — Environmental Setup

pip install torch transformers scikit-learn pandas numpy matplotlib scipy nltk

Step 2 — Download the Datasets Download the five CSV files from the lab GitHub repository:

https://github.com/ideas-labo/ISE/tree/main/lab1

Place all five CSV files in the same directory as br_classification_full.py.

Step 3 — Verify Configuration Open br_classification_full.py and confirm the following settings at the top of the file match exactly:

PROJECTS = ['tensorflow', 'pytorch', 'keras', 'incubator-mxnet', 'caffe']

REPEAT = 30

TEST_SIZE = 0.30

BERT_MODEL = 'distilbert-base-uncased'

BERT_EPOCHS = 3

BERT_MAX_LEN = 256

BERT_BATCH = 16

BERT_LR = 2e-5

Step 4 — Run the Experiment

cd ISE-CW

python3 br_classification_full.py

Step 5 — Verify Outputs Upon completion the following files should be present in results/:

```
results/
├── results_summary.csv
├── cross_project_results.csv
├── tensorflow_raw_scores.csv
├── pytorch_raw_scores.csv
├── keras_raw_scores.csv
├── incubator-mxnet_raw_scores.csv
├── caffe_raw_scores.csv
└── plots/
    ├── tensorflow_boxplots.png
    ├── tensorflow_histograms.png
    ├── pytorch_boxplots.png
    ├── pytorch_histograms.png
    ├── keras_boxplots.png
    ├── keras_histograms.png
    ├── incubator-mxnet_boxplots.png
    ├── incubator-mxnet_histograms.png
    ├── caffe_boxplots.png
    ├── caffe_histograms.png
    └── all_projects_f1_boxplot.png
```

Step 6 — Verify Key Results The following median F1 values should be reproduced:

Dataset	Baseline Median F1	DistilBERT Median F1
Tensorflow	0.5557	0.8513
Pytorch	0.4657	0.7224
Keras	0.5319	0.7247
MXNet	0.5069	0.7388
Caffe	0.4691	0.5693